

2.13 KEY TERMS, REVIEW QUESTIONS, AND PROBLEMS

Key Terms

batch processing batch system execution context interrupt job job control language kernel kernel mode mean time to failure (MTTF) mean time to repair (MTTR) memory management microkernel monitor monolithic kernel	multiprogrammed batch system multiprogramming multitasking multithreading nucleus operating system physical address privileged instruction process process state real address resident monitor round-robin	scheduling serial processing symmetric multiprocessing task thread time sharing time-sharing system time slicing uniprogramming user mode virtual address virtual machine virtual memory
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Review Questions

- 2.1. What are three objectives of an OS design?
- 2.2. What is the kernel of an OS?
- 2.3. What is multiprogramming?
- 2.4. What is a process?
- 2.5. How is the execution context of a process used by the OS?
- 2.6. List and briefly explain five storage management responsibilities of a typical OS.
- 2.7. Explain the distinction between a real address and a virtual address.
- 2.8. Describe the round-robin scheduling technique.
- 2.9. Explain the difference between a monolithic kernel and a microkernel.
- 2.10. What is multithreading?
- 2.11. List the key design issues for an SMP operating system.

Problems

- 2.1. Suppose that we have a multiprogrammed computer in which each job has identical characteristics. In one computation period, T , for a job, half the time is spent in I/O and the other half in processor activity. Each job runs for a total of N periods. Assume that a simple round-robin scheduling is used, and that I/O operations can overlap with processor operation. Define the following quantities:
 - Turnaround time = actual time to complete a job
 - Throughput = average number of jobs completed per time period T
 - Processor utilization = percentage of time that the processor is active (not waiting)

Compute these quantities for one, two, and four simultaneous jobs, assuming that the period T is distributed in each of the following ways:

- a. I/O first half, processor second half
 - b. I/O first and fourth quarters, processor second and third quarter
- 2.2. An I/O-bound program is one that, if run alone, would spend more time waiting for I/O than using the processor. A processor-bound program is the opposite. Suppose a short-term scheduling algorithm favors those programs that have used little processor time in the recent past. Explain why this algorithm favors I/O-bound programs and yet does not permanently deny processor time to processor-bound programs.
- 2.3. Contrast the scheduling policies you might use when trying to optimize a time-sharing system with those you would use to optimize a **multiprogrammed batch** system.
- 2.4. What is the purpose of system calls, and how do system calls relate to the OS and to the concept of dual-mode (kernel-mode and user-mode) operation?
- 2.5. In IBM's mainframe OS, OS/390, one of the major modules in the kernel is the System Resource Manager. This module is responsible for the allocation of resources among address spaces (processes). The SRM gives OS/390 a degree of sophistication unique among operating systems. No other mainframe OS, and certainly no other type of OS, can match the functions performed by SRM. The concept of resource includes processor, real memory, and I/O channels. SRM accumulates statistics pertaining to utilization of processor, channel, and various key data structures. Its purpose is to provide optimum performance based on performance monitoring and analysis. The installation sets forth various performance objectives, and these serve as guidance to the SRM, which dynamically modifies installation and job performance characteristics based on system utilization. In turn, the SRM provides reports that enable the trained operator to refine the configuration and parameter settings to improve user service.

This problem concerns one example of SRM activity. Real memory is divided into equal-sized blocks called frames, of which there may be many thousands. Each frame can hold a block of virtual memory referred to as a page. SRM receives control approximately 20 times per second and inspects each and every page frame. If the page has not been referenced or changed, a counter is incremented by 1. Over time, SRM averages these numbers to determine the average number of seconds that a page frame in the system goes untouched. What might be the purpose of this and what action might SRM take?

- 2.6. A multiprocessor with eight processors has 20 attached tape drives. There is a large number of jobs submitted to the system that each require a maximum of four tape drives to complete execution. Assume that each job starts running with only three tape drives for a long period before requiring the fourth tape drive for a short period toward the end of its operation. Also assume an endless supply of such jobs.
- a. Assume the scheduler in the OS will not start a job unless there are four tape drives available. When a job is started, four drives are assigned immediately and are not released until the job finishes. What is the maximum number of jobs that can be in progress at once? What are the maximum and minimum number of tape drives that may be left idle as a result of this policy?
 - b. Suggest an alternative policy to improve tape drive utilization and at the same time avoid system deadlock. What is the maximum number of jobs that can be in progress at once? What are the bounds on the number of idling tape drives?